



Homework 8: Breit-Wigner resonance and Gadolinium.

This question will explore transitions from an initial state $|k_i\rangle$ to a final state $|k_f\rangle$ that proceed through an intermediate resonance $|R\rangle$, which is an unstable state with complex energy $\mathcal{E}_R$.

This is a continuation of HW 7.

Consider the non-relativistic scattering of two particles through an intermediate state $|R\rangle$ of H_0 (R for “resonance”). Working in the center-of-mass frame, we can treat the problem as a single particle of reduced mass μ scattering off a potential V that allows for the resonance $|R\rangle$ to be created or destroyed. The scattering states are plane waves in a periodic box of volume L^3 ,

$$\langle \mathbf{r} | \mathbf{k} \rangle = \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{L^{3/2}}, \quad (1)$$

so that $\langle \mathbf{k}' | \mathbf{k} \rangle = \delta_{\mathbf{k}'\mathbf{k}}$ and the density of states in k -space is $L^3/(2\pi)^3$. The resonance $|R\rangle$ is a discrete, unit-normalized eigenstate of H_0 . Assume for simplicity that $\langle R | V | \mathbf{k} \rangle$ depends only on $|\mathbf{k}|$ and not on $\hat{\mathbf{k}}$ (the “s-wave” nomenclature comes from the partial wave expansion of the scattering amplitude, where $\ell = 0$ corresponds to the spherically symmetric partial wave (s-wave), $\ell = 1$ to the p-wave, $\ell = 2$ to the d-wave, and so on).

From the second-order Dyson series, the transition amplitude from $|\mathbf{k}_i\rangle$ to $|\mathbf{k}_f\rangle$ through the resonance is

$$T_{fi}^{(\text{box})} = \frac{\langle \mathbf{k}_f | V | R \rangle \langle R | V | \mathbf{k}_i \rangle}{E - \mathcal{E}_R}, \quad (2)$$

where $\mathcal{E}_R = E_R + \Delta E_R - i\Gamma/2$ is the complex energy from HW 7 part (d). We define the normalized matrix element

$$\bar{V}_{R\mathbf{k}} \equiv L^{3/2} \langle R | V | \mathbf{k} \rangle. \quad (3)$$

This is just the matrix element taken with the plane wave $e^{i\mathbf{k}\cdot\mathbf{r}}$ rather than $e^{i\mathbf{k}\cdot\mathbf{r}}/L^{3/2}$.

a) Show that the box-normalized matrix element squared is

$$|T_{fi}^{(\text{box})}|^2 = \frac{|\bar{V}_{R\mathbf{k}_f}|^2 |\bar{V}_{R\mathbf{k}_i}|^2}{L^6 [(E - \tilde{E}_R)^2 + \Gamma^2/4]}, \quad (4)$$

where $\tilde{E}_R \equiv E_R + \Delta E_R$.

b) A resonance $|R\rangle$ may decay into several distinct sets of final states, called “decay channels.” For example, the excited nucleus $^{17}\text{O}^*$ can be produced and can decay through channels such as $n + ^{16}\text{O}$ (channel a) or $p + ^{16}\text{N}$ (channel b). The process $n + ^{16}\text{O} \rightarrow ^{17}\text{O}^* \rightarrow p + ^{16}\text{N}$ is then a scattering from channel a to channel b through the resonance. Each channel has its own coupling to $|R\rangle$ and its own “partial width” $\Gamma_a, \Gamma_b, \dots$, defined as the decay rate of $|R\rangle$ into that channel alone (for example, Γ_a is the decay rate of $^{17}\text{O}^*$ into $n + ^{16}\text{O}$ and Γ_b is the decay rate into $p + ^{16}\text{N}$). The total width is

$$\Gamma = \Gamma_a + \Gamma_b + \dots \quad (5)$$

We may also define a branching ratio

$$\text{BR}_a = \frac{\Gamma_a}{\Gamma}, \quad (6)$$

where it should be clear that $\sum_a \text{BR}_a = 1$. Clearly, our example b is inelastic and ought to have a different partial width Γ_b than the elastic example a . In what follows, however, we will assume

that all channels are elastic, so that the initial and final momenta are the same, $k_i = k_f = k$.¹

The decay rate of $|R\rangle$ into all states of a particular channel is still given by Fermi's Golden Rule,

$$\Gamma_a = 2\pi \sum_{\mathbf{k}} |\langle \mathbf{k} | V | R \rangle|^2 \delta(E_R - E_k) \longrightarrow \frac{\mu k_R}{\pi} |\bar{V}_R|_a^2, \quad (7)$$

where $E_k = k^2/(2\mu)$ and $k_R = \sqrt{2\mu E_R}$. We keep the subscript a to indicate that the matrix element is for the channel a . For another channel b , Γ_b takes the same form with the appropriate matrix element $|\bar{V}_R|_b^2$. Show the correspondence in Eq. (7) by replacing the sum over $\mathbf{k}$ with an integral and using the density of states in k -space. Assume that the matrix element is independent of the direction of $\mathbf{k}$, so that the integral over d^3k reduces to an integral over dk , which you can evaluate using the delta function. Then express $|\bar{V}_{Rk_i}|^2$ and $|\bar{V}_{Rk_f}|^2$ in terms of the partial widths Γ_a and Γ_b .

(Hint: use the property of the delta function $\delta(f(k)) = \sum_{k_0} \delta(k - k_0)/|f'(k_0)|$ with k_0 the roots of f .)

c) Apply Fermi's Golden Rule for the scattering rate into a solid angle $d\Omega_f$:

$$dw = 2\pi |T_{fi}^{(\text{box})}|^2 \rho_f(E) d\Omega_f, \quad \rho_f(E) = \frac{L^3 \mu k_f}{(2\pi)^3}, \quad (8)$$

where $\rho_f(E)$ is the density of final states per unit energy per unit solid angle. Dividing by the incident flux $\Phi = v_i/L^3 = k_i/(\mu L^3)$, derive the **differential** cross section, verifying that all factors of L cancel:

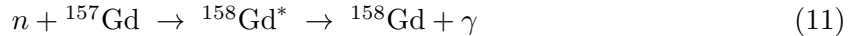
$$\frac{d\sigma}{d\Omega_f} = \frac{\mu^2}{4\pi^2} \frac{|\bar{V}_{Rk_f}|^2 |\bar{V}_{Rk_i}|^2}{(E - \tilde{E}_R)^2 + \Gamma^2/4}. \quad (9)$$

Then integrate over $d\Omega_f$ (using the s-wave isotropy and $k_f = k_i = k$ for elastic scattering) and express the result in terms of the partial widths from part (b). Show that the total cross section is

$$\sigma(E) = \frac{\pi}{k^2} \frac{\Gamma_a \Gamma_b}{(E - \tilde{E}_R)^2 + \Gamma^2/4} \quad (10)$$

where $\Gamma = \Gamma_a + \Gamma_b + \dots$ is the total width.

(d) Gadolinium-157 has the largest thermal neutron capture cross section of any stable nuclide, a property exploited in nuclear reactor control rods and in gadolinium-doped water Cherenkov neutrino detectors (such as the Super-Kamiokande experiment). The process



proceeds through a compound nuclear resonance at $E_R = 0.0314$ eV. The entrance channel (a) is the neutron and the exit channel (b) is the photon. The resonance parameters are²

$$\Gamma_n^{\text{eff}} = g\Gamma_n^0 \sqrt{E_R/\text{eV}} = 0.590 \text{ meV} \times \sqrt{0.0314} \approx 0.105 \text{ meV}, \quad \Gamma_\gamma = 106 \text{ meV}, \quad \Gamma \approx \Gamma_\gamma. \quad (12)$$

¹Technically, for any channel the resonance energy E_R is the same, but the momentum k is different because of the different reduced mass μ in each channel. We ignore this subtlety and assume that k is the same for all channels, though it can be easily generalized by noting that if a given channel has a threshold energy $Q_{a,b}$, then $k_{a,b} = \sqrt{2\mu_{a,b}(E_R - Q_{a,b})}$. For the process to be kinematically allowed, we need $E_R > Q_{a,b}$, so that $k_{a,b}$ is real.

²The effective neutron width $\Gamma_n^{\text{eff}} \equiv g_J \Gamma_n$ absorbs the spin statistical factor $g_J = (2J+1)/[(2s+1)(2I+1)] = 5/8$, which counts the fraction of incident spin channels that can form the $J^\pi = 2^-$ compound state from an s -wave neutron ($s = 1/2$) on a ${}^{157}\text{Gd}$ target ($I = 3/2$). This detail is needed for unpolarized beams, but the Breit-Wigner formula from Eq. (10) applies as written with $\Gamma_a = \Gamma_n^{\text{eff}}$. The tabulated value $g\Gamma_n^0 = 0.590$ meV is the *reduced* width at a reference energy of 1 eV; the physical width at the resonance energy is $\Gamma_n^{\text{eff}}(E_R) = g\Gamma_n^0 \sqrt{E_R/1 \text{ eV}}$.

In the Breit–Wigner formula (10), use $\Gamma_a = \Gamma_n^{\text{eff}}$ and $\Gamma_b = \Gamma_\gamma$. The reduced mass is $\mu \approx m_n(1 - 1/(A + 1)) \approx 933.6$ MeV, with $A = 157$.

Plot $\sigma(E)$ in the range $E = 0.001$ – 1 eV. Overlay the unitarity bound $4\pi/k^2$ and the geometric cross section πR_{nuc}^2 with $R_{\text{nuc}} = 1.2 A^{1/3}$ fm. Use logarithmic axes.

(Bonus question) In a water Cherenkov detector, an inverse beta decay event ($\bar{\nu}_e + p \rightarrow e^+ + n$) produces a neutron that thermalizes and is subsequently captured. Capture on hydrogen ($n + p \rightarrow d + \gamma$, $E_\gamma = 2.2$ MeV) produces a single low-energy photon that is difficult to detect. Capture on gadolinium produces a γ -ray cascade totaling ~ 8 MeV, which is much easier to identify. This motivates doping the water with a small amount of gadolinium.

Consider water doped with 0.01% Gd by mass (the approximate loading of Super-Kamiokande). The number density of ^{157}Gd atoms in this solution is $n_{157} \approx 6.0 \times 10^{16} \text{ cm}^{-3}$.³ The number density of hydrogen atoms in water is $n_H \approx 6.7 \times 10^{22} \text{ cm}^{-3}$, and the thermal neutron capture cross section on hydrogen is $\sigma_H \approx 0.33$ barn.

- (i) Compute the macroscopic cross sections $\Sigma_{\text{Gd}} = n_{157} \sigma_{\text{Gd}}$ and $\Sigma_H = n_H \sigma_H$ at thermal energies. Use the single-resonance σ_{Gd} from part (d).
- (ii) The mean free path for neutron capture is $\lambda = 1/(\Sigma_H + \Sigma_{\text{Gd}})$. Compute λ with and without Gd doping.
- (iii) What fraction of neutrons are captured by ^{157}Gd rather than by hydrogen? Why is this advantageous for detecting reactor antineutrinos?

³This follows from $n_{157} = f_{\text{Gd}} \eta_{157} \rho_w N_A / A_{\text{Gd}}$, where $f_{\text{Gd}} = 10^{-4}$ is the Gd mass fraction, $\eta_{157} = 15.65\%$ is the isotopic abundance, $\rho_w \approx 1 \text{ g/cm}^3$, N_A is Avogadro's number, and $A_{\text{Gd}} \approx 157 \text{ g/mol}$.

Solutions



This question guides us through the derivation of the Breit-Wigner formula for the cross section of a scattering process that proceeds through an intermediate resonance.

a) The box-normalized matrix element between the resonance and a scattering state is

$$\langle \mathbf{k} | V | R \rangle = \frac{1}{L^{3/2}} \int d^3 \mathbf{x} e^{-i \mathbf{k} \cdot \mathbf{x}} V(\mathbf{x}) \psi_R(\mathbf{x}) = \frac{\bar{V}_{R\mathbf{k}}^*}{L^{3/2}}, \quad (13)$$

where we inserted a position basis completeness relation and used the resonance state wavefunction $\psi_R(\mathbf{x}) = \langle \mathbf{x} | R \rangle$. Note that $\langle R | V | \mathbf{k} \rangle = \bar{V}_{R\mathbf{k}}/L^{3/2}$. Therefore, from Eq. (2),

$$T_{fi}^{(\text{box})} = \frac{\bar{V}_{R\mathbf{k}_f}^* \bar{V}_{R\mathbf{k}_i}}{L^3 (E - \mathcal{E}_R)}, \quad (14)$$

and

$$|T_{fi}^{(\text{box})}|^2 = \frac{|\bar{V}_{R\mathbf{k}_f}|^2 |\bar{V}_{R\mathbf{k}_i}|^2}{L^6 [(E - \tilde{E}_R)^2 + \Gamma^2/4]}, \quad (15)$$

where $\tilde{E}_R = E_R + \Delta E_R$ and we used $|\Delta|^2 = (\Re \Delta)^2 + (\Im \Delta)^2$.

f) Let us first convert the sum over box-normalized states to a continuum integral. We will need the density of states in momentum space:

$$\sum_{\mathbf{k} \in a} \rightarrow \frac{L^3}{(2\pi)^3} \int d^3 k. \quad (16)$$

$$\Gamma_a = 2\pi \frac{L^3}{(2\pi)^3} \int d^3 k \frac{|\bar{V}_{R\mathbf{k}}|^2}{L^3} \delta(E_R - E_k). \quad (17)$$

The factors of L^3 cancel! This is great news and it means we can take the $L \rightarrow \infty$ limit without worrying about the normalization of the states. The s-wave assumption

$$|\bar{V}_{R\mathbf{k}}|^2 = |\bar{V}_R|^2, \quad (18)$$

amounts to saying that the matrix element is isotropic in $\hat{\mathbf{k}}$ and depends only on $k = |\mathbf{k}|$ (the angular dependence is trivial). Therefore, we can integrate the angular variables trivially, $d^3 k = 4\pi k^2 dk$. Then, we use the root of the delta function $k_R = \sqrt{2\mu E_R}$ in the provided hint to get

$$\Gamma_a = \frac{2\pi}{(2\pi)^3} 4\pi |\bar{V}_R|^2 \int_0^\infty dk k^2 \frac{\mu}{k_R} \delta(k - k_R) = \frac{|\bar{V}_R|^2 \mu k_R}{\pi}. \quad (19)$$

b) The transition rate into solid angle $d\Omega_f$ is

$$dw = 2\pi |T_{fi}^{(\text{box})}|^2 \rho_f d\Omega_f = 2\pi \frac{|\bar{V}_{R\mathbf{k}_f}|^2 |\bar{V}_{R\mathbf{k}_i}|^2}{L^6 [(E - \tilde{E}_R)^2 + \Gamma^2/4]} \frac{L^3 \mu k_f}{(2\pi)^3} d\Omega_f. \quad (20)$$

The cross section is $d\sigma = dw/\Phi$ with incident flux $\Phi = v_i/L^3 = k_i/(\mu L^3)$:

$$\frac{d\sigma}{d\Omega_f} = \frac{dw}{\Phi d\Omega_f} = 2\pi \frac{|\bar{V}_{R\mathbf{k}_f}|^2 |\bar{V}_{R\mathbf{k}_i}|^2}{L^6 [(E - \tilde{E}_R)^2 + \Gamma^2/4]} \frac{L^3 \mu k_f}{(2\pi)^3} \frac{\mu L^3}{k_i}. \quad (21)$$

Indeed, all six powers of L cancel, giving

$$\frac{d\sigma}{d\Omega_f} = \frac{\mu^2 k_f}{4\pi^2 k_i} \frac{|\bar{V}_{R\mathbf{k}_f}|^2 |\bar{V}_{R\mathbf{k}_i}|^2}{(E - \tilde{E}_R)^2 + \Gamma^2/4}. \quad (22)$$

For elastic scattering, $k_f = k_i = k$, and the s-wave matrix elements are isotropic. Integrating over the solid angle $d\Omega_f$ gives a factor of 4π :

$$\sigma = \frac{\mu^2}{\pi} \frac{|\bar{V}_R|_b^2 |\bar{V}_R|_a^2}{(E - \tilde{E}_R)^2 + \Gamma^2/4}. \quad (23)$$

We now use the result from Eq. (19) to express $|\bar{V}_R|^2 = \pi\Gamma/(\mu k)$ and evaluate near resonance where $k \approx k_R$,

$$\sigma = \frac{\mu^2}{\pi} \frac{\pi\Gamma_b}{\mu k} \frac{\pi\Gamma_a}{\mu k} \frac{1}{(E - \tilde{E}_R)^2 + \Gamma^2/4} = \frac{\pi}{k^2} \frac{\Gamma_a \Gamma_b}{(E - \tilde{E}_R)^2 + \Gamma^2/4}. \quad (24)$$

This is the Breit–Wigner formula for the cross section of a scattering process that proceeds through an intermediate resonance. The step of evaluating near resonance is justified by the fact that the cross section is sharply peaked around $E \simeq \tilde{E}_R$ when Γ is small.

When evaluating the denominator at $E \simeq \tilde{E}_R$, we have the narrow-width approximation (NWA). As long as the width of the resonance is narrow, the NWA is well justified. What the NWA does for us is simplify the full scattering process into a two-step one: first the resonance is produced with a rate Γ_a , and then it decays with a rate Γ_b . Note that since $|c_i(t)|^2 = e^{-\Gamma t}$, the lifetime of the resonance is $\tau = 1/\Gamma$. Therefore, a narrow resonance corresponds to a long-lived state and a broad resonance to a short-lived one.

For purely elastic scattering ($\Gamma = \Gamma_a$, only one channel), the peak cross section is

$$\sigma_{\max} = \sigma(\tilde{E}_R) = \frac{\pi}{k^2} \frac{\Gamma_a^2}{\Gamma^2/4} = \frac{4\pi}{k^2}. \quad (25)$$

Later in the course, we will see that this value *saturates* the s-wave unitarity bound $\sigma_{\ell=0} \leq 4\pi/k^2$. In other words, this is the maximum cross section that an s-wave amplitude can have without violating conservation of probability.

For different entry and exit channels with $\Gamma = \Gamma_a + \Gamma_b + \dots$, the NWA cross section at resonance is

$$\sigma_{\max} = \frac{\pi}{k^2} \frac{\Gamma_a \Gamma_b}{\Gamma^2/4} = \frac{4\pi}{k^2} \frac{\Gamma_a \Gamma_b}{\Gamma^2} = \frac{4\pi}{k^2} \text{BR}_a \text{BR}_b. \quad (26)$$

(d) We evaluate the Breit–Wigner formula (10) with parameters $E_R = 0.0314$ eV, $\Gamma_a = \Gamma_n^{\text{eff}} = 0.105$ meV, $\Gamma_b = \Gamma_\gamma = 106$ meV, $\Gamma \approx 106$ meV, and $\mu = 933.6$ MeV.

The resonance momentum is ($\hbar c = 197.3$ MeV fm)

$$k_R = \frac{\sqrt{2\mu E_R}}{\hbar c} = \frac{\sqrt{2 \times 933.6 \times 3.14 \times 10^{-8}}}{197.3} \text{ fm}^{-1} \approx 4 \times 10^{-5} \text{ fm}^{-1}, \quad (27)$$

which corresponds to a de Broglie wavelength $\lambda = 1/k_R \sim 2.6$ nm. Note that $\lambda \gg R_{\text{nuc}} \sim \mathcal{O}(\text{fm})$. That is, the neutron's quantum wavelength is four thousand times the nuclear radius. This already tells us that the cross section will be set by the wavelength, not the geometry.

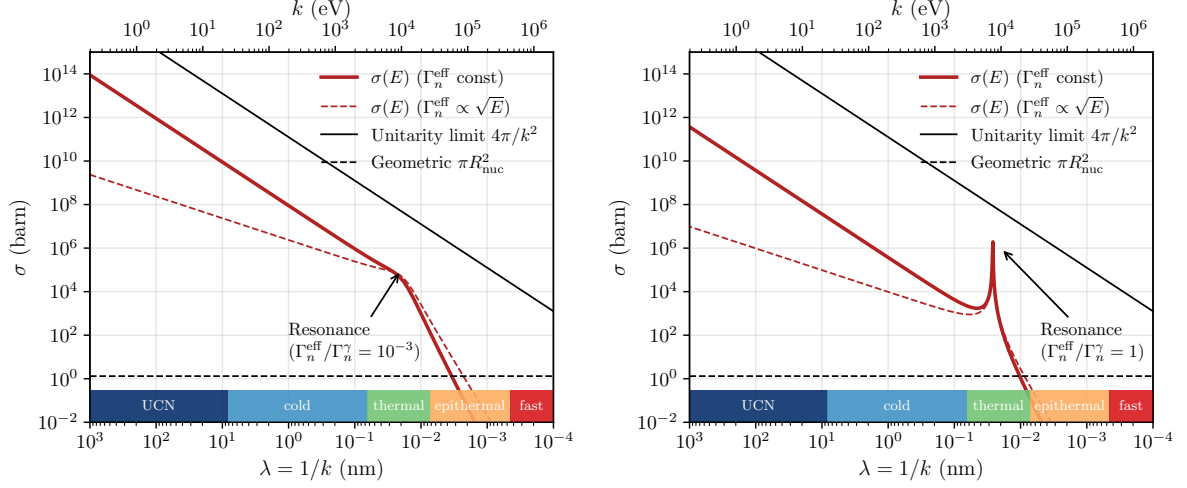


Figure 1: The neutron capture cross section on ^{157}Gd as a function of the neutron wavelength (momentum k , top axis). Left panel shows the actual case with $\Gamma_n^{\text{eff}}/\Gamma_\gamma \approx 10^{-3}$, while the right panel shows a hypothetical case where we decrease the exit channel such that $\Gamma_\gamma = \Gamma_n^{\text{eff}}$. The red line shows the total cross section from the Breit–Wigner formula (solid for constant width and dashed for energy-dependent width⁵), while the black solid and dashed lines show the unitarity bound and geometric cross section, respectively.

(i) The unitarity bound at resonance is

$$\frac{4\pi}{k_R^2} = \frac{4\pi}{(4 \times 10^{-5})^2} \text{ fm}^2 \approx 8 \times 10^7 \text{ barn}. \quad (28)$$



A barn is a unit of area/cross section commonly used in nuclear and particle physics, defined as $1 \text{ barn} = 10^{-28} \text{ m}^2 = 100 \text{ fm}^2$.

This is actually an enormous value, but the peak cross section is far below the unitarity bound because $\Gamma_n^{\text{eff}} \ll \Gamma_\gamma$ (or, equivalently, $\text{BR}_a \ll 1$):

$$\sigma_{\text{max}} = \frac{4\pi}{k^2} \frac{\Gamma_a \Gamma_b}{\Gamma^2} \approx \frac{4\pi}{k^2} \frac{\Gamma_n^{\text{eff}}}{\Gamma_\gamma} = 8.35 \times 10^7 \times \frac{1.05 \times 10^{-4}}{0.106} \text{ barn} \approx 8.2 \times 10^4 \text{ barn}. \quad (29)$$

The geometric cross section is

$$\pi R_{\text{nuc}}^2 = \pi (1.2 \times 157^{1/3})^2 \text{ fm}^2 \approx 1.3 \text{ barn}. \quad (30)$$

This is clearly undershooting the actual value, so this scattering process is controlled by the neutron's de Broglie wavelength rather than the geometric size of the nucleus.